

A Sim-to-Real Integration Pipeline for Training and Deployment of Chunk-Based VLA Manipulation Policies

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Abstract: A sim-to-real pipeline for deploying expert trajectories in open-loop for VLA training and performing closed-loop inference with the trained policy on real-world hardware.

Code: : https://gitlab.isir.upmc.fr/kappel/sim2real_public_chunk_control/, **Contact :** kappel@isir.upmc.fr and grislain@isir.upmc.fr

Introduction

Challenges in VLA sim2real integration

- Real-world teleoperation demonstrations are expensive.
- Sim-to-real gap is rarely quantified → conflates policy quality, perception, and physical mismatch into a single number.

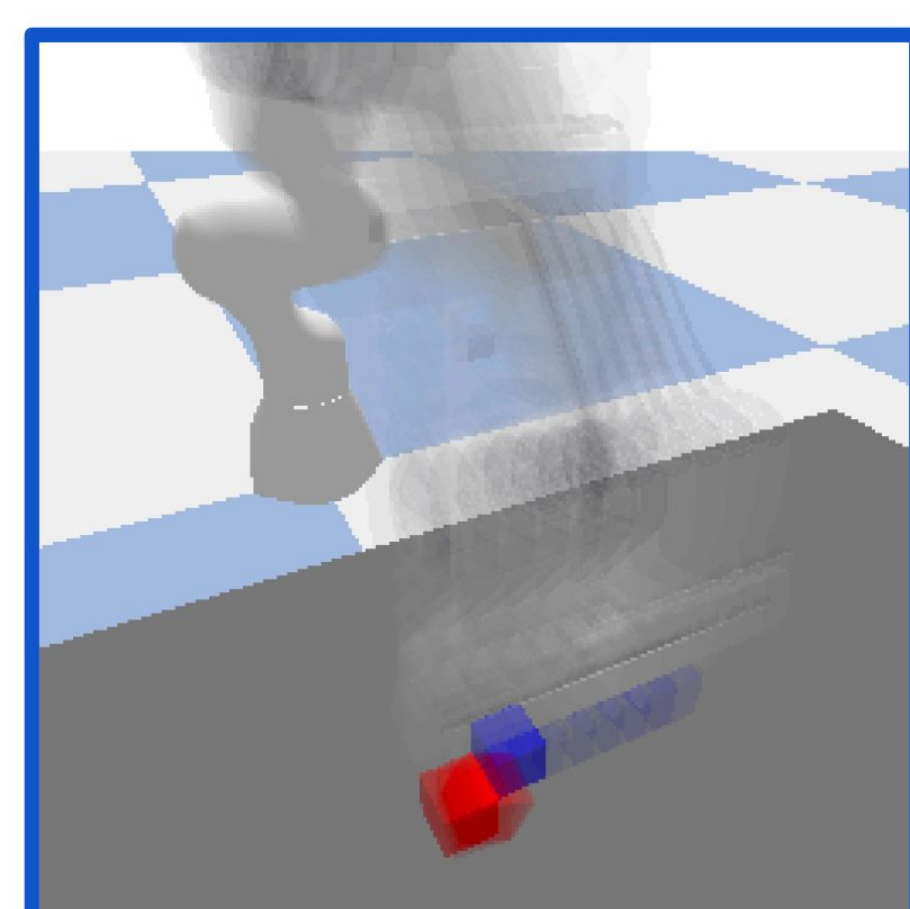


How can I train and deploy chunk-based VLA manipulation policies?

Methods

Sim2Real Setup

- Paired simulated and real world setup: Franka FR3 arm and objects to grasp (URDF cubes).
- Visual Observations: Orbec Femto Bolt RGB.
- The real initial scene configuration is reproduced using the robot itself as a Cartesian pointer.



Action Chunk Representation and Execution

- π_{VLA} Policy produces an action chunk of fixed size

$$a_{SimChunk}^t = (a_{sim,1}^t, a_{sim,2}^t, \dots, a_{sim,k}^t)$$

$$a_{sim,i}^t = (\Delta eef_i^t, g_i^t) \in \mathcal{A}$$

$$(\Delta x_i^t, \Delta y_i^t, \Delta z_i^t, \delta q_{1,i}^t, \delta q_{2,i}^t, \delta q_{3,i}^t, \delta q_{4,i}^t) \quad g \in \{0, 1\}$$

One variation of the end-effector frame (eef) is a change in translation and rotation of the eef

One gripper configuration

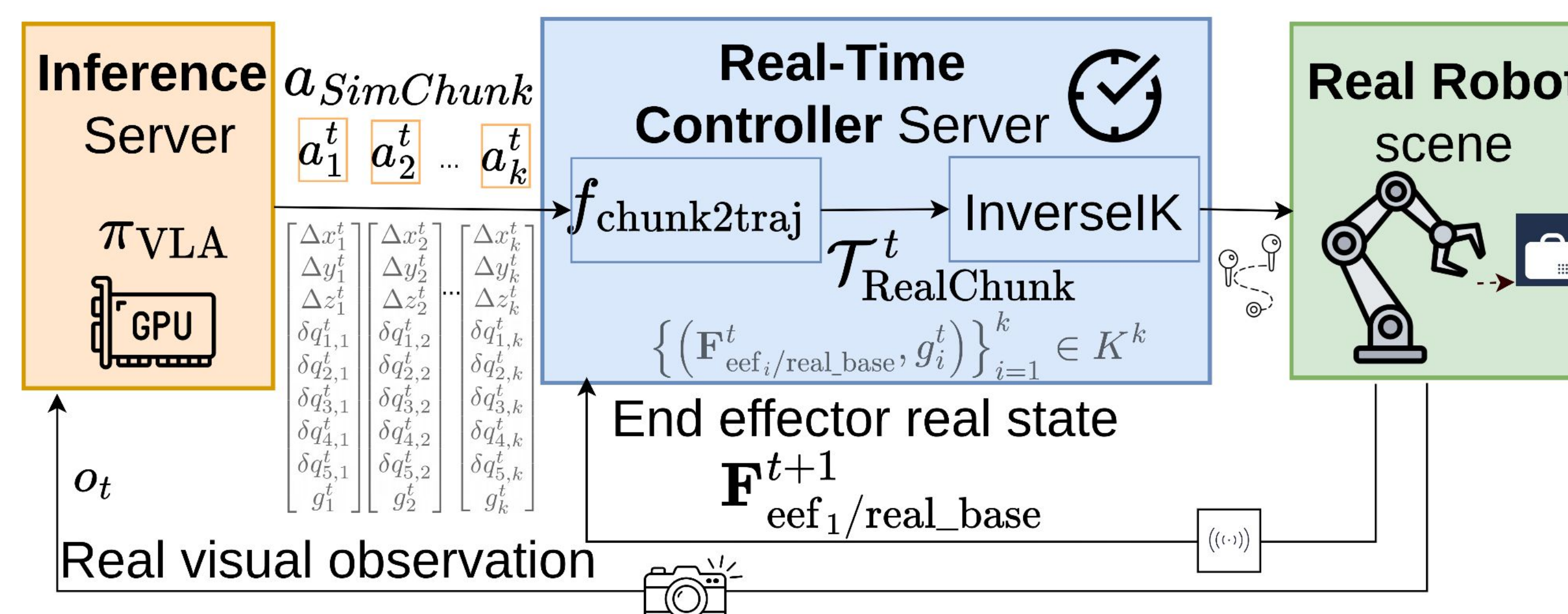
- Each action chunk is converted into a sequence of target states for the real robot's end-effector + gripper state.

$$f_{chunk2traj} : SE(3) \times \mathcal{A}^k \rightarrow \mathcal{K}^k$$

$$(F_{eef_1/real_base}^t, a_{SimChunk}^t) \mapsto \mathcal{T}_{RealChunk}^t$$

- Use of a MoveGroup path planner

Expert and Inference sim2real Deployment



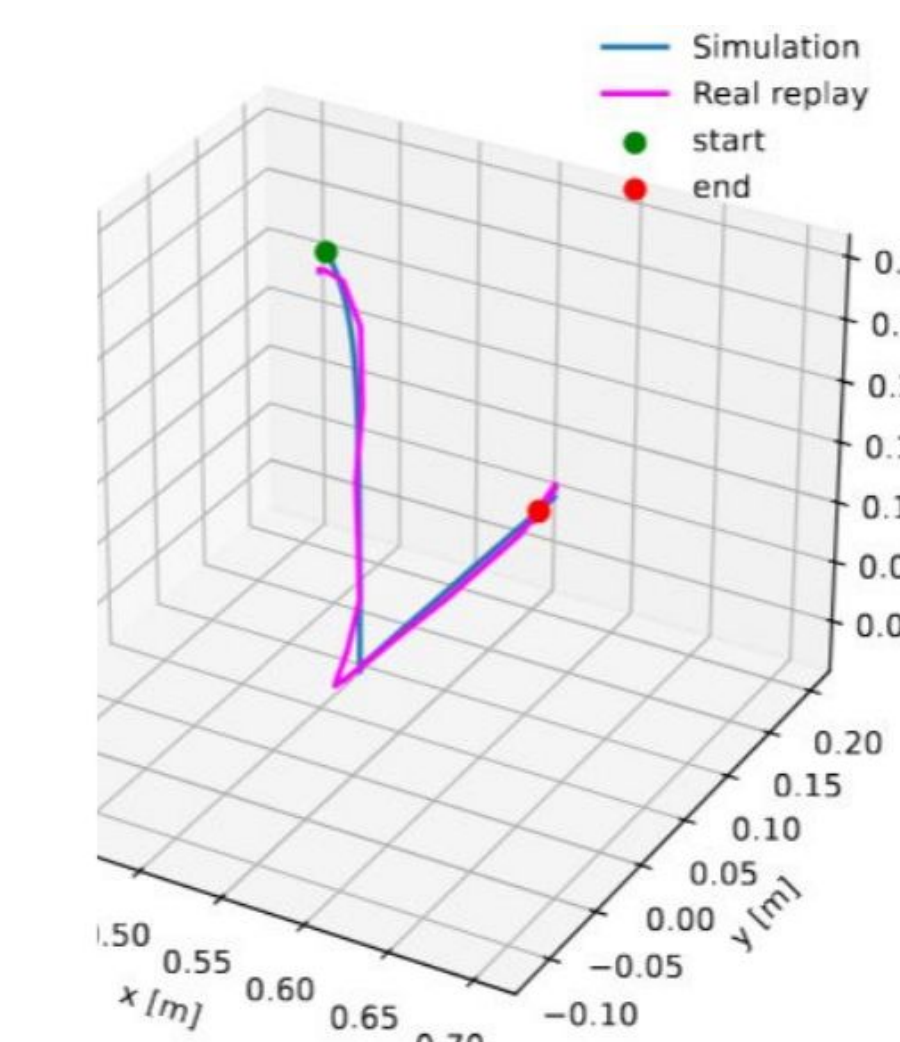
- Expert Data Collection: The execution of all the expert task chunk is open-loop

$$\forall t \in \llbracket 1, m-1 \rrbracket, F_{eef_1/real_base}^{t+1} = F_{eef_k/real_base}^t$$

- Closed loop inference deployment: Accomplish the task by inferring actions based on visual observations.
- A full task deployment relies on sequential executions of *Inference* and *Command* ROS 2 custom *Actions*.

Experiments & Results

- Collection of 200 real-world data for VLA training, action chunk size $k = 8$.
- Average eef path deviation & Fréchet distance



Metric	Reaching	Contact	Full
Fréchet distance (cm)	2.77 ± 0.08	3.30 ± 0.05	3.30 ± 0.05
Average eef path deviation (cm)	0.90 ± 0.03	0.96 ± 0.02	0.93 ± 0.02
Task success rate	—	—	100%

- Real controller consistently tracks the commanded path & deviations occur while the robot **interacts** with the cube.
- 100% transfer for expert replay.
- 30% success rate for inference using trained VLA (ORCHID [1]).